



Lakshya JEE 2.0 (2024)

Indefinite Integration

DPP-06

1. $\int [\sin \log x + \cos \log x] dx$ is equal to

- (1) $\sin(\log x) + \cos(\log x) + C$
- (2) $x \sin(\log x) + C$
- (3) $x \cos(\log x) + C$
- (4) $x \tan x + C$

2. $\int \sqrt{2x^2 - x + 1} dx$

3. Evaluate $\int (2x - 4)\sqrt{4 + 3x - x^2} dx$

4. $\int e^{3x} \sin 2x dx$ equals

- (1) $\frac{e^{3x}}{13} (3 \cos 2x + 2 \sin 2x) + C$
- (2) $\frac{e^{3x}}{13} (3 \sin 2x - 2 \cos 2x) + C$
- (3) $\frac{e^{3x}}{13} (-3 \sin 2x - 2 \cos 2x) + C$
- (4) $\frac{e^{3x}}{13} (3 \sin 2x + 2 \cos 2x) + C$

5. $\int \frac{x^2 + 1}{(x-1)^2(x+3)} dx$ equals

- (1) $e^{x-1} + \frac{1}{4} \cdot \frac{1}{(x-1)} + \tan^{-1}(x+3) + \log x + C$
- (2) $\frac{3}{8} \log |x-1| - \frac{1}{2} \cdot \frac{1}{(x-1)} + C$
- (3) $\frac{3}{8} \log |x-1| - \frac{1}{2} \cdot \frac{1}{(x-1)} + \frac{5}{8} \log |x+3| + C$
- (4) $\frac{1}{3} \tan^{-1}(x+1) + \log |x+3| + C$

6. If $\int \frac{2x+3}{(x-1)(x^2+1)} dx = -a \left(\log \left\{ \frac{(x-1)^2}{x^2+1} \right\} - \frac{1}{2} \tan^{-1} x + C, x > 1 \right.$ where C is any arbitrary

constant, then the value of 'a' is

- (1) $\frac{5}{4}$
- (2) $-\frac{5}{3}$
- (3) $-\frac{5}{6}$
- (4) $-\frac{5}{4}$

7. If $\int \frac{f(x)}{x^3-1} dx$, where $f(x)$ is a polynomial of degree

2 in x such that $f(0) = f(1) = 3f(2) = -3$ and

$$\int \frac{f(x)}{x^3-1} dx = -\log |x-1| + \log |x^2+x+1|$$

$+ \frac{m}{\sqrt{n}} \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + C$. Then $(2m+n)$ is

- (1) 3
- (2) 5
- (3) 7
- (4) 9

8. If $\int \frac{2x^2+3}{(x^2-1)(x^2+4)} dx = a \ln \left(\frac{x-1}{x+1} \right) +$

$b \tan^{-1} \frac{x}{2} + C$, then values of a and b are

respectively.

- (1) $\frac{1}{2}, \frac{1}{3}$
- (2) 1, 1
- (3) $\frac{1}{2}, \frac{1}{2}$
- (4) $\frac{1}{3}, \frac{1}{3}$

9. The value of integral $\int e^{\frac{x^2+1}{x}} \left(2x^2 - \frac{1}{x} + 1 \right) dx$ is equal to (where C is the constant of integration)

- (1) $e^{\frac{x^2+1}{x}} + C$
- (2) $x^2 e^{\frac{x^2+1}{x} + C}$
- (3) $x e^{\frac{x^2+1}{x}} + C$
- (4) $x e^{x^2} + C$

10. The value of integral

$$\int e^x \left[\left(\frac{2 \tan x}{1 + \tan x} \right) + \cot^2 \left(x + \frac{\pi}{4} \right) \right] dx \text{ is equal to:}$$

- (1) $e^x \tan \left(\frac{\pi}{4} - x \right) + C$
- (2) $e^x \tan \left(x - \frac{\pi}{4} \right) + C$
- (3) $e^x \tan \left(\frac{3\pi}{4} - x \right) + C$
- (4) $e^x \tan \left(x - \frac{3\pi}{4} \right) + C$

11. $\int e^{\tan \theta} \sec \theta - \sin \theta) d\theta$ equals

- (1) $-e^{\tan \theta} \sin \theta + C$
- (2) $e^{\tan \theta} \sin \theta + C$
- (3) $e^{\tan \theta} \sec \theta + C$
- (4) $e^{\tan \theta} \cos \theta + C$

12. The graph of the antiderivative of $f(x) = xe^{\frac{x}{2}}$ pass through (0, 3) then the value of $g(2) - f(0)$ is

- (1) 1
- (2) 3
- (3) 5
- (4) 7

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (2)

2. 7/16

3. $-\frac{2}{3}(4+3x-x^2)^{3/2} - \frac{1}{2}$

$$\left[\left(x - \frac{3}{2} \right) \sqrt{4+3x-x^2} + \frac{25}{4} \sin^{-1} \left(\frac{2x-3}{5} \right) \right] + C$$

4. (2)

5. (3)

6. (4)

7. (3)

8. (3)

9. (3)

10. (2)

11. (4)

12. (4)



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>